

Assignment 2 : Project Report

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Course: Intro To SE

Instructor: Prof. M. Umer Haroon

University: NUCES Peshawar Campus

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1. Project Overview

This project focuses on improving a legacy-style C++ loan/EMI calculator. The original codebase had several issues: - Lack of input validation - Hardcoded values for calculations - Risk of integer overflow for large loans or long tenures - Minimal organization, all logic was in a single file

The objective of this assignment was to: - Identify and fix bugs - Refactor the code for better maintainability - Implement unit tests to ensure correct functionality - Document the project using Doxygen

2. Tasks Completed

A. Git and Project Setup

- Forked the starter repository and created a separate dev branch
- Added .gitignore to exclude build artifacts
- Updated README.md with instructions to build and run the project
- Maintained commit history for every major change

B. Bug Fixes and Refactoring

- Input Validation: Added checks to ensure principal, interest rate, and tenure are valid numbers; negative values are rejected
- Integer Overflow Prevention: Used long double for calculations to prevent overflow in large loans or long-term tenures
- Code Refactoring: Split the monolithic main.cpp into separate files:
 - o Loan.cpp / Loan.h for loan calculation logic
 - o Utils.cpp for helper functions
 - o main.cpp for user input and output handling
- Improved readability and maintainable output formatting

C. Unit Testing

- Implemented GoogleTest unit tests to verify:
 1. Normal EMI calculations
 2. Invalid input handling (negative numbers or zero tenure)
 3. Large tenure calculations without overflow
- Ensured all critical functionalities pass tests reliably

D. Documentation

- Added Doxygen comments to all functions and class definitions
 - Generated HTML documentation to visually represent class structure, member functions, and relationships
 - Organized docs under docs/ folder for easy access
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3. Screenshots and Test Output

```
vinod-kumar@vinod-kumar-Latitude-7420:~/projects/legacy-calc-2009$ g++ -std=c++17 src/main.cpp src/Loan.cpp -o LoanCLI
./LoanCLI
Enter loan amount: 5000
Enter interest rate: 34
Enter years: 2

Total amount: 13400
Total interest: 3400
Monthly payment: 558
vinod-kumar@vinod-kumar-Latitude-7420:~/projects/legacy-calc-2009$
```

```
vinod-kumar@vinod-kumar-Latitude-7420:~/projects/legacy-calc-2009$ g++ -std=c++17 -Isrc src/Loan.cpp test/test_emi.cpp -lgtest -lgtest_main -pthread -o LoanTest
./LoanTest
[=====] Running 3 tests from 3 test suites.
[-----] Global test environment set-up.
[-----] 1 test from EMI_Normal
[ RUN      ] EMI_Normal.Basic
[       OK ] EMI_Normal.Basic (0 ms)
[-----] 1 test from EMI_Normal (0 ms total)

[-----] 1 test from InvalidInput
[ RUN      ] InvalidInput.NegativeValues
[       OK ] InvalidInput.NegativeValues (0 ms)
[-----] 1 test from InvalidInput (0 ms total)

[-----] 1 test from LargeTenure
[ RUN      ] LargeTenure.NoOverflow
[       OK ] LargeTenure.NoOverflow (0 ms)
[-----] 1 test from LargeTenure (0 ms total)

[-----] Global test environment tear-down
[=====] 3 tests from 3 test suites ran. (0 ms total)
[ PASSED  ] 3 tests.
vinod-kumar@vinod-kumar-Latitude-7420:~/projects/legacy-calc-2009$
```

4. Git Commit History

Discuss and review the changes in this comparison with others. [Learn about pull requests](#)

Create pull request

6 commits

116 files changed

1 contributor

Commits on Nov 26, 2025

Fix Bugs and Implement Testing

Vinod-kumar-malhi committed 3 hours ago

0de0e82 <>

Bug Fixing In Calculator and Add Unit Testing

Vinod-kumar-malhi committed 3 hours ago

54385e2 <>

Adding Testing

Vinod-kumar-malhi committed 3 hours ago

d5c3fad <>

Creating Doxygenfile

Vinod-kumar-malhi committed 2 hours ago

f29e823 <>

Creating Doxygenfile

Vinod-kumar-malhi committed 2 hours ago

74c2563 <>

Creating DoxygenFile

Vinod-kumar-malhi committed 1 hour ago

059ac36 <>

Showing 116 changed files with 9,727 additions and 1,332 deletions.

SplitUnified